Supplementary Material for "Multi-Resolution Global Horizontal Irradiance Forecasting at Electric-Vehicle Manufacturing Locations: TimesNet and Dilated Recurrent Networks"

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1. Complete Reference-Model Benchmark

This supplement compares TimesNet and DilatedRNN with all evaluated reference models at each cumulative forecast horizon. Table S1 reports the complete macro-MAE comparison, and Figure S1 shows how model rankings vary across horizons. Figure S2 summarizes variability across random seeds for all learned models.

Table S1: Macro test MAE (W/m^2) at cumulative forecast horizons. Bold marks the lowest unrounded MAE in each row. Daily and monthly learned-model results use five-seed ensemble forecasts; hourly results use one seed. The Seeds column refers to learned models; baselines are deterministic. NA denotes hourly climatology, which was not evaluated. Complete metrics and exact-lead results are recorded in the evaluation CSV files.

Task	Horizon	Climatology	DilatedRNN	LSTM	Persistence	Seasonal naïve	TimesNet	XGBoost	Seeds
Hourly 336–72	24 h	NA	50.73	50.91	52.58	65.91	49.39	**48.37**	1
Hourly 336–72	48 h	NA	52.15	52.79	56.16	65.92	52.16	**49.81**	1
Hourly 336–72	72 h	NA	52.96	53.39	58.11	65.90	52.48	**50.50**	1
Daily 365–30	1 d	45.74	43.27	42.04	43.87	58.90	43.20	**41.58**	5
Daily 365–30	3 d	45.76	43.80	43.14	50.17	58.98	43.63	**43.09**	5
Daily 365–30	7 d	45.75	44.08	**43.69**	53.77	59.00	43.95	43.71	5
Daily 365–30	14 d	45.64	44.11	44.00	56.41	58.91	44.09	**43.96**	5
Daily 365–30	30 d	45.51	**43.85**	44.18	60.28	58.78	44.08	43.93	5
Monthly 12–1	1 mo	**12.57**	13.01	14.27	35.09	16.94	13.89	14.82	5
Monthly 12–6	1 mo	**13.61**	14.89	15.74	33.19	16.50	14.86	14.73	5
Monthly 12–6	3 mo	**13.08**	14.33	15.33	61.45	17.06	14.49	14.93	5
Monthly 12–6	6 mo	**13.03**	14.37	14.76	91.89	18.00	14.46	14.72	5

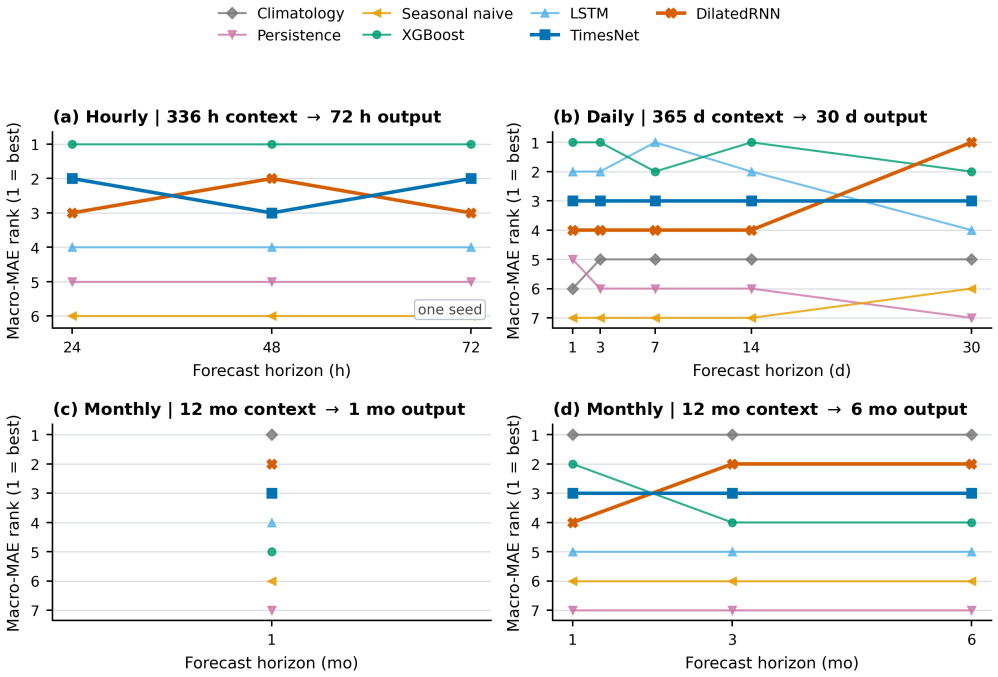


Figure S1: Macro-MAE rankings for all models at cumulative forecast horizons. Rank 1 indicates the lowest MAE within each task and horizon. Hourly climatology was not evaluated.

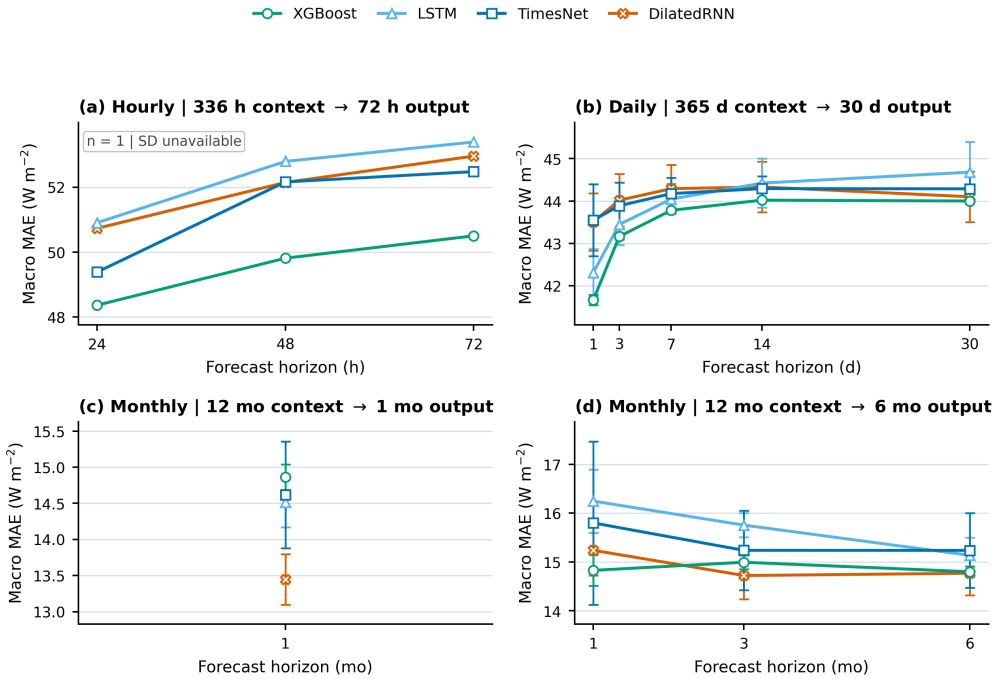


Figure S2: Variability across random seeds for all learned models. Error bars show one sample standard deviation; hourly results use one seed and therefore have no error bars.

2. Detailed Forecasting Algorithms

The three blocks below provide additional preparation and evaluation details for the protocol and architecture pseudocode in the main article. They focus on the matched TimesNet–DilatedRNN comparison. Algorithm S1 defines data preparation, Algorithm S2 specifies selection and refitting, and Algorithm S3 records post-processing and evaluation.

Algorithm S1 Leakage-controlled data preparation

Require: Site series $\{y_{\ell,t}\}$; task (r, C, H); fixed cutoffs
Ensure: Aligned initial training, validation, refit, and test windows
for each site ℓ **do**
  Parse, sort, and verify a complete nonnegative GHI series
  if r is monthly **then**
    Aggregate daily GHI by civil-month arithmetic mean
  end if
  Assign the fixed identifier ℓ
  Fit s_ℓ^{sel} using observations before 2023
  Fit s_ℓ^{refit} using observations before 2024
  for each declared partition P **do**
    Form contexts $y_{\ell,t-C:t-1}$ with site ID ℓ
    Retain only full targets $y_{\ell,t:t+H-1}$ lying in P
    Assert the expected origin count and chronological boundaries
  end for
end for
Concatenate sites and sort each partition by origin and site identifier

Algorithm S2 TimesNet–DilatedRNN selection and direct forecasting

Require: Block-A partitions; fixed seeds and model settings
Ensure: Validation and test forecasts on common origins in W/m^2
if hourly extension **then**
  Verify hashes and load the official TimesNet seed-42 forecasts
  $\mathcal{M} \leftarrow \{\text{DilatedRNN}\}$; $\mathcal{S} \leftarrow \{42\}$
else
  $\mathcal{M} \leftarrow \{\text{TimesNet}, \text{DilatedRNN}\}$
  $\mathcal{S} \leftarrow \{11, 23, 42, 67, 89\}$
end if
Scale initial training and validation with s_ℓ^{sel}
Scale refit and test with s_ℓ^{refit}
for each $(m, s) \in \mathcal{M} \times \mathcal{S}$ **do**
  Initialize m with seed s and fit on the initial training partition using Adam
  Select $E^\star$ by early stopping on normalized 2023 validation MSE
  Retain the validation forecast from the $E^\star$ checkpoint
  Reinitialize m with seed s
  Refit on the pre-2024 partition for exactly $E^\star$ epochs
  Forecast all H targets at every 2024 origin with fixed weights
  Invert s_ℓ^{sel} for validation and s_ℓ^{refit} for test
  Retain unbounded validation and test forecasts for this seed in W/m^2
end for
For daily/monthly models, average the five raw seed forecasts
Retain seed and ensemble forecasts separately, before applying the zero floor

Algorithm S3 Post-processing, metrics, and contextual analysis

Require: Block-B forecasts; targets; declared horizons; context metadata
Ensure: Site-level, macro, horizon-wise, and contextual evidence
for each newly computed seed or ensemble forecast **do**
    Apply $\widehat{y} \leftarrow \max(\widehat{y}, 0)$ in W/m^2
    if the task is hourly **then**
        Compute solar elevation at each hourly interval's midpoint
        Set $\widehat{y} = 0$ wherever solar elevation is nonpositive
    end if
end for
Verify common site, origin, target timestamp, and lead keys
for each site and declared horizon **do**
    Compute metrics for the cumulative prefix and the exact lead
end for
Average site metrics with equal weight for each site
Form TimesNet–DilatedRNN differences paired by site
Attach climate and adverse-day labels only after errors have been computed

3. Supporting Design and Literature Tables

Table S2 summarizes the four forecasting tasks and their origin counts. Tables S3, S4, and S5 cover three groups of studies, respectively: intra-hour and short hourly forecasting; day-ahead and direct multi-step forecasting; and monthly, multi-climate, and very-long-horizon forecasting.

Table S2: Fixed multi-resolution forecasting tasks.

Task	Context → output	Evaluated horizons	Origins/site A/V/R/T	Role
Hourly	336 h → 72 h	24, 48, 72 h	1,445/363/1,810/363	Official protocol; DilatedRNN seed-42 extension
Daily	365 d → 30 d	1, 3, 7, 14, 30 d	1,067/336/1,432/337	Primary matched task
Monthly-1	12 mo → 1 mo	1 mo	36/12/48/12	Primary monthly task
Monthly-6	12 mo → 6 mo	1, 3, 6 mo	31/7/43/7	Exploratory multi-month task

A/V/R/T denote initial training (adjustment), 2023 validation, pre-2024 refit, and 2024 test. Every count is the number of complete forecast origins per site. Reported multi-step horizons are evaluated both as cumulative prefixes and exact leads.

Table S3: Representative forecasting studies at intra-hour and short hourly horizons. The studies use different targets and test protocols, so their results are not directly ranked.

Study	Target and scope	Inputs and processing	Predictor	Horizon and evaluation design
Chow et al. [1]	GHI and cloud conditions at the UC San Diego test bed	30-s total-sky images, clear-sky library, cloud map, shadows, and motion vectors	Image-based physical nowcast	Cloud nowcasts up to 5 min; six pyranometers on four partly cloudy days; qualitative GHI assessment
Guariso et al. [2]	Hourly GHI at an Italian station, with transfer to other sites	Autoregressive irradiance windows; no decomposition	Feed-forward and LSTM networks	1–12 h; recursive, horizon-specific, and multi-output strategies; persistence and clear-sky references
Huang et al. [3]	Hourly irradiance at Denver, Clark, and Folsom	Irradiance history and meteorology, including next-hour temperature, humidity, and wind-speed forecasts; wavelet-packet decomposition	WPD–CNN–LSTM–MLP hybrid	1 h ahead; comparison with individual networks and climatology–persistence

Appendix A. Reproducibility Specification

This appendix specifies the configuration used to reproduce the matched experiments. The executable protocol binds task definitions, input and source-code hashes, software versions, model settings, and random seeds into a contract hash. Cached work is reused only when that complete hash matches.

Table S4: Representative day-ahead and direct multi-step solar-forecasting studies.

Study	Target and scope	Inputs and processing	Predictor	Horizon and evaluation design
Cheng et al. [4]	Daytime hourly irradiance in Taiwan under tropical and subtropical marine conditions	Previous days of irradiance and selected temperature, humidity, and rainfall observations or forecasts	One-dimensional CNN, LSTM, skip concatenation, and dense output	Day-ahead through 7 days ahead; limited-data deployment experiment
Voyant et al. [5]	Solar-radiation series from sites in multiple climates	Little or no training history; statistical structure of the target	Five naïve references (including a second-order autoregressive model) and their forecast combination	Direct multi-step benchmark evaluated with several complementary metrics
Zhao et al. [6]	Hourly solar radiation	ICEEMDAN-decomposed solar-radiation sequences combined with meteorological data	ICEEMDAN–TimesNet	Week-ahead multi-step prediction; comparison with alternative multi-step models

Table S5: Representative monthly, multi-climate, and very-long-horizon forecasting studies.

Study	Target and scope	Inputs and processing	Predictor	Horizon and evaluation design
Ozoegwu [7]	Monthly-mean daily global solar radiation at selected locations	Radiation history; lagged month numbers in NARX and current month number in the hybrid	NAR, NARX, and hybrid ANN	Monthly-scale forecasting; comparison among the three neural formulations
Zhu et al. [8]	Solar irradiance in a photovoltaic-greenhouse setting	Multiple meteorological-factor combinations and training-set lengths	TimesNet, Informer, Transformer, and Pyraformer	Nine horizons from 10 min to 24 h; seasonal and full-climate analyses
Cabral et al. [9]	GHI at 40 locations spanning multiple climate classes	GHI-only enhanced pattern-recognition pipeline	GPT-Neo and contemporary neural competitors	Very-long-term multi-year prediction; location and aggregate errors plus computational assessment

Abbreviations in Tables S3–S5: ANN, artificial neural network; CNN, convolutional neural network; GHI, global horizontal irradiance; ICEEMDAN, improved complete ensemble empirical mode decomposition with adaptive noise; LSTM, long short-term memory; MLP, multilayer perceptron; NAR, nonlinear autoregressive; NARX, nonlinear autoregressive with exogenous inputs; WPD, wavelet packet decomposition.

Appendix A.1. Temporal partitions

Table S6 gives inclusive forecast-origin boundaries. The target cutoff is checked separately, so a multi-step window is retained only when every target lies on or before that date. The initial training partition is labeled "adjustment" in the execution artifacts.

Table S6: Complete chronological partitions and expected origins per site.

Task	Partition	Origin range	Declared target cutoff	*N*
Hourly 336–72	Initial training	2019-01-15–2022-12-29	2022-12-31	1,445
	Validation	2023-01-01–2023-12-29	2023-12-31	363
	Refit	2019-01-15–2023-12-29	2023-12-31	1,810
	Test	2024-01-01–2024-12-28	2024-12-31	363
Daily 365–30	Initial training	2020-01-01–2022-12-02	2022-12-31	1,067
	Validation	2023-01-01–2023-12-02	2023-12-31	336
	Refit	2020-01-01–2023-12-02	2023-12-31	1,432
	Test	2024-01-01–2024-12-02	2024-12-31	337
Monthly 12–1	Initial training	2020-01-31–2022-12-31	2022-12-31	36
	Validation	2023-01-31–2023-12-31	2023-12-31	12
	Refit	2020-01-31–2023-12-31	2023-12-31	48
	Test	2024-01-31–2024-12-31	2024-12-31	12
Monthly 12–6	Initial training	2020-01-31–2022-07-31	2022-12-31	31
	Validation	2023-01-31–2023-07-31	2023-12-31	7
	Refit	2020-01-31–2023-07-31	2023-12-31	43
	Test	2024-01-31–2024-07-31	2024-12-31	7

The cutoff column records the calendar boundary enforced by the window constructor, not necessarily the last observed target timestamp. In particular, the hourly test's last origin is 2024-12-28 00:00 and its 72 hourly targets end at 2024-12-30 23:00 in the fixed local-time index; UTC timestamps for western sites may extend into 31 December.

The 2019 observations preceding the first daily and monthly origins belong only to the causal context. Selection scales use all available observations before 2023-01-01, and refit scales use all available observations before 2024-01-

01. Each scale is site-specific and records its observation count, minimum, maximum, and effective range.

These partitions are rolling-origin evaluations. Parameters and scale values are fixed before the relevant evaluation year, while the context advances and can contain earlier observations from that year after they become available. No parameter update occurs within validation or test.

Appendix A.2. Model and optimization settings

The common learned-model input is an array of shape (batch, context length) containing only scaled GHI, with one integer site identifier per example. The target is a vector of continuous GHI values covering the declared forecast horizon. No learned model receives latitude, longitude, clock variables, solar elevation, climate, cloud, precipitation, or a previously predicted target. Table S7 records the fixed model, optimizer, seed, and ensemble settings.

Table S7: Fixed settings for all learned models.

Item	**Value**
Seeds	11, 23, 42, 67, 89 for daily/monthly; 42 for the official hourly protocol and its DilatedRNN extension
Neural objective	Mean squared error on site-wise min–max normalized GHI
Optimizer	Adam; learning rate 10^{-3}; weight decay 10^{-5}
Batch/selection	Hourly and daily: batch 128, 30 epochs maximum, patience 5; monthly: batch 32, 300 epochs maximum, patience 30
Determinism	Python, NumPy, and PyTorch generators fixed; deterministic PyTorch algorithms requested; one PyTorch CPU thread
Site representation	Learned embedding of width 8 in TimesNet, DilatedRNN, and LSTM; one-hot vector for XGBoost
TimesNet	$d_{\text{model}} = 8$, $d_{\text{ff}} = 16$, one TimesBlock, top-$k = 3$, two Inception kernels, dropout 0.1, direct linear temporal head; sample-specific FFT period selection and per-window standardization with $\epsilon = 10^{-5}$
DilatedRNN	Vanilla tanh-RNN cells with recurrent dilations $(1, 2, 4)$, 16 units in each layer, dense width 16, dropout 0, direct output head
LSTM	One recurrent layer, hidden width 32, direct output head
XGBoost	Squared-error objective, 150 trees, depth 3, learning rate 0.05, subsample 0.8, column subsample 0.8, histogram tree method, one thread; multi-output-tree strategy when the output length exceeds one
Refit rule	Fresh initialization under the same seed and exactly the epoch count selected using 2023 validation data; XGBoost is independently fitted with its fixed budget
Ensemble	Arithmetic mean of five seed forecasts for daily/monthly learned models; no hourly ensemble

Hourly daily-profile persistence repeats the last 24 observed hourly values cyclically over the forecast horizon. The annual seasonal naïve reference uses the value at the same local hour in the preceding year. For daily and monthly tasks, persistence repeats the last observed aggregate throughout the forecast horizon, whereas annual seasonal naïve uses the same civil day or month from the preceding year, with 29 February mapped to 28 February when needed. Daily and monthly climatology uses a fixed mean for the target calendar day or month, computed from observations before the relevant selection or test cutoff. A global mean computed before the cutoff is used when no observations match the calendar day or month. All baseline and learned forecasts are clipped to zero if negative. Hourly outputs are also set to zero when solar elevation is nonpositive, evaluated at minute 30 of each interval.

Appendix A.3. Evaluation outputs and reproducibility records

The retained corrected archive is identified as `avaliacao_multirresolucao_corrigida_v2` under the project's `resultados/` directory. Its task directories are `hourly_72_extension`, `daily_30`, `monthly_1`, and `monthly_6`. This identifier specifies the source of the reported comparisons; it does not imply that the full forecast bundle is currently available in the public repository.

For every declared horizon, the pipeline writes cumulative-prefix and exact-lead metrics. It stores raw and post-processed forecasts, metrics by site, equal-weight macro metrics, site-paired TimesNet–DilatedRNN differences, selected epoch counts, training histories, and the two sets of scales fitted before the respective cutoffs. Daily and monthly

task directories additionally store variability across the five seeds. The principal machine-readable files across task directories are:

- `contrato_execucao.json` and `hiperparametros_e_metodo.json`;
- `protocolo_temporal.csv`, `escalas_minmax_pre_corte.csv`, and `epocas_selecionadas.csv`;
- `previsoes_validacao.csv.gz` and `previsoes_teste.csv.gz`;
- `metricas_por_localidade.csv`, `metricas_macro.csv`, and `variabilidade_sementes.csv` (daily/monthly only);
- `comparacao_pareada_localidades.csv`, `comparacao_pareada_resumo.csv`, and `manifesto_artefatos.json`.

The contract records Python, NumPy, pandas, PyTorch, scikit-learn, XGBoost, and joblib versions together with SHA-256 hashes of source code and input files. Smoke mode reduces sites, origins, seeds, epochs, and trees and is marked nonpublishable in every status artifact.

Appendix A.4. Geographic and meteorological provenance

The geographic CSV retains all factory-coordinate provenance, raster-cell row and column, cell-center coordinates, sampling distance, class code and descriptions, confidence, source citation, file paths, and the MD5 values in Table S8. MD5 is used only as a reproducibility checksum, not as a security mechanism.

Table S8: Integrity records for the external contextual layer.

Asset	**Role and retained checksum**
Beck present 1-km class raster	Site assignment; MD5 `fe8f23532bfb59231a072398f0d9a7cb`
Beck present 1-km confidence raster	Assignment confidence; MD5 `a87f331e406e4ef02114e1350eb2dd01`
Beck present 0.5-degree raster	Americas-map background only; MD5 `78d09475fa4b0dbb004b6178cba47b4a`
Beck class legend	Codes, descriptions, and RGB colors; MD5 `f0632ca1b3b475d3de5a143c4ee1928f`
`contexto_geografico.csv`	Auditable sampled table; SHA-256 `80b6e003a326d6142df03766c15a5dbe28c7c1dc0be3c195df07a60b764f7f41`
`contexto_meteorologico_2024.csv`	Post-hoc NASA POWER context; SHA-256 `7fbf97076193f2467e0092cf6b0d2c9a00ea19c84f354b28f1736f5bba95f4b5`

The climate source is the 1980–2016 nominal 1-km Köppen–Geiger product of Beck et al. [10]. Class confidence records agreement among the 12 ensemble maps used to derive that classification. The meteorological artifact contains 3,660 location–day rows in local solar time and is accompanied by a JSON manifest that records the NASA POWER query URL and SHA-256 of every cached response [11]. These data are used only to contextualize forecast errors after evaluation. A day is classified as adverse if precipitation is at least 1 mm/day or cloud amount is at least 80%. The manifest explicitly records that these variables are not model inputs.

Appendix B. Additional Result Tables

Table S9 reports the means and sample standard deviations of macro-MAE across seeds at each cumulative horizon, as shown in the variability plots. Horizons are expressed in hours, days, or months according to the task. Table S10 describes the adverse-weather example selected independently of either model's errors.

Table S9: Macro test MAE (W/m^2) across seeds at cumulative forecast horizons. Daily and monthly entries report the mean of individual-seed MAEs over five seeds, with the sample standard deviation in parentheses. Hourly results use one seed (n = 1).

Task	Cumulative horizon	DilatedRNN	LSTM	TimesNet	XGBoost
Hourly 336–72	24 h	50.73 (n = 1)	50.91 (n = 1)	49.39 (n = 1)	48.37 (n = 1)
Hourly 336–72	48 h	52.15 (n = 1)	52.79 (n = 1)	52.16 (n = 1)	49.81 (n = 1)
Hourly 336–72	72 h	52.96 (n = 1)	53.39 (n = 1)	52.48 (n = 1)	50.50 (n = 1)
Daily 365–30	1 d	43.51 (0.68)	42.31 (0.55)	43.55 (0.85)	41.67 (0.12)
Daily 365–30	3 d	44.02 (0.62)	43.45 (0.48)	43.89 (0.54)	43.16 (0.07)
Daily 365–30	7 d	44.29 (0.57)	44.05 (0.36)	44.17 (0.37)	43.78 (0.06)
Daily 365–30	14 d	44.33 (0.60)	44.42 (0.58)	44.29 (0.29)	44.02 (0.03)
Daily 365–30	30 d	44.10 (0.60)	44.68 (0.71)	44.29 (0.29)	44.00 (0.02)
Monthly 12–1	1 mo	13.45 (0.35)	14.51 (0.34)	14.62 (0.74)	14.86 (0.17)
Monthly 12–6	1 mo	15.24 (0.51)	16.25 (0.65)	15.80 (1.67)	14.83 (0.31)
Monthly 12–6	3 mo	14.72 (0.49)	15.75 (0.24)	15.24 (0.82)	14.99 (0.14)
Monthly 12–6	6 mo	14.77 (0.45)	15.14 (0.36)	15.24 (0.76)	14.80 (0.11)

Table S10: Meteorological case selected independently of model errors.

Field	**Value**
Site	Hyundai Metaplant Georgia
Local date	2024-08-05
Corrected precipitation (mm/day)	82.66
Cloud amount (%)	99.47
Precipitation ≥ 1 mm/day	Yes
Cloud amount ≥ 80%	Yes
Adverse-day criterion satisfied	Yes
Interpretation	Descriptive meteorological example selected independently of errors.

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